

# Skills Summary

## The Rational Root Theorem: Finding Every Zero

Use this reference while completing your worksheet or when studying.

The theorem does not hand you the zeros — it hands you a *short list of suspects*. Testing, dividing, and finishing the leftover quadratic is where the zeros actually come from.

### 1. Build the list and test a candidate (verify-a-root)

**Theorem:** if a polynomial with integer coefficients has a rational zero  $\frac{p}{q}$  in lowest terms, then  $p$  divides the **constant term** and  $q$  divides the **leading coefficient**.

Candidates =  $\pm \frac{\text{divisors of the constant}}{\text{divisors of the leading coefficient}}$ . A candidate is a zero exactly when substituting it gives 0 (equivalently, synthetic division leaves remainder 0).

**Example:** Is  $x = 3$  a zero of  $f(x) = x^3 - 2x^2 - 5x + 6$ ?

**Step 1.** The constant is 6 and the leading coefficient is 1, so 3 is on the candidate list — test it by substitution.

**Step 2.**  $f(3) = 27 - 18 - 15 + 6$   **$f(3) = 0$** , so  $x = 3$  is a zero and  $(x - 3)$  is a factor.

**Try it:** Is  $x = -2$  a zero of  $x^3 + x^2 - 4x - 4$ ?

yes —  **$0 = f(-2)$**

### 2. Find all rational zeros (all-rational-zeros)

**Routine:** list candidates  $\rightarrow$  test until the remainder is 0  $\rightarrow$  divide that factor out  $\rightarrow$  factor or solve what is left.

A leading coefficient other than 1 puts fractions on the list: for  $2x^3 + \dots + 3$  the candidates are  $\pm 1, \pm 3, \pm \frac{1}{2}, \pm \frac{3}{2}$ . Try small integers first — they are usually the ones that work.

**Example:** All zeros of  $2x^3 + 3x^2 - 8x + 3$ .

**Step 1.** Leading coefficient 2 and constant 3, so test the short integers on the list before the halves.

**Step 2.**  $x = 1$  gives  $2 + 3 - 8 + 3 = 0$ ; dividing leaves  $2x^2 + 5x - 3 = (2x - 1)(x + 3)$ .  
 **$x = 1, x = \frac{1}{2}, x = -3$**

**Try it:** Find all zeros of  $3x^3 - 4x^2 - 5x + 2$ .

yes —  **$x = 1, x = \frac{2}{3}, x = -\frac{1}{2}$**

### 3. Factor completely (factor-completely)

**Link:** a zero  $x = c$  means the factor  $(x - c)$ ; a zero  $x = \frac{p}{q}$  means the integer factor  $(qx - p)$ . Keep dividing until what remains is a quadratic you can factor or a quadratic with no real zeros. *Completely* means no factor can be broken down further over the integers.

**Example:** Factor  $x^3 - 6x^2 + 11x - 6$ .

**Step 1.** Constant  $-6$  and leading coefficient  $1$ : test the divisors of  $6$  and divide out the first one that works.

**Step 2.**  $x = 1$  is a zero; division leaves  $x^2 - 5x + 6 = (x - 2)(x - 3)$ .  $(x - 1)(x - 2)(x - 3)$

**Try it:** Factor  $x^3 + 3x^2 - x - 3$  completely.

$(1 + x)(1 - x)(3 + x)$  ✓

#### 4. When the leftovers are not rational (mixed-zeros)

**Key idea:** the theorem finds only *rational* zeros. After dividing one out, finish the quadratic with the quadratic formula — that is where irrational zeros ( $a \pm \sqrt{b}$ ) and non-real zeros ( $a \pm bi$ ) come from.

A degree- $n$  polynomial has  $n$  zeros counted with multiplicity, so keep going until you have  $n$  of them.

**Example:** All zeros of  $x^3 - 5x^2 + x + 7$ .

**Step 1.** Only  $\pm 1, \pm 7$  are candidates, so at most one zero will come from the list — expect to finish with the quadratic formula.

**Step 2.**  $x = -1$  works; division leaves  $x^2 - 6x + 7$ , so  $x = \frac{6 \pm \sqrt{36 - 28}}{2} = 3 \pm \sqrt{2}$ .  
 $x = -1, x = 3 \pm \sqrt{2}$

**Try it:** Find all zeros of  $x^3 - 2x^2 + 9x - 18$ , including non-real ones.

$18 - 9x = x^2, 2x = x^2, 18 = x^2$  ✓

**(!) Watch out:** An empty candidate list result means “no *rational* zero,” never “no zero.” Put  $p$  over the constant’s divisors and  $q$  over the leading coefficient’s — reversing them is the most common slip. And a zero of  $\frac{1}{2}$  pairs with the factor  $(2x - 1)$ .